

PROJECT APPRAISAL

Module 8

The Distribution of Project Net Benefits



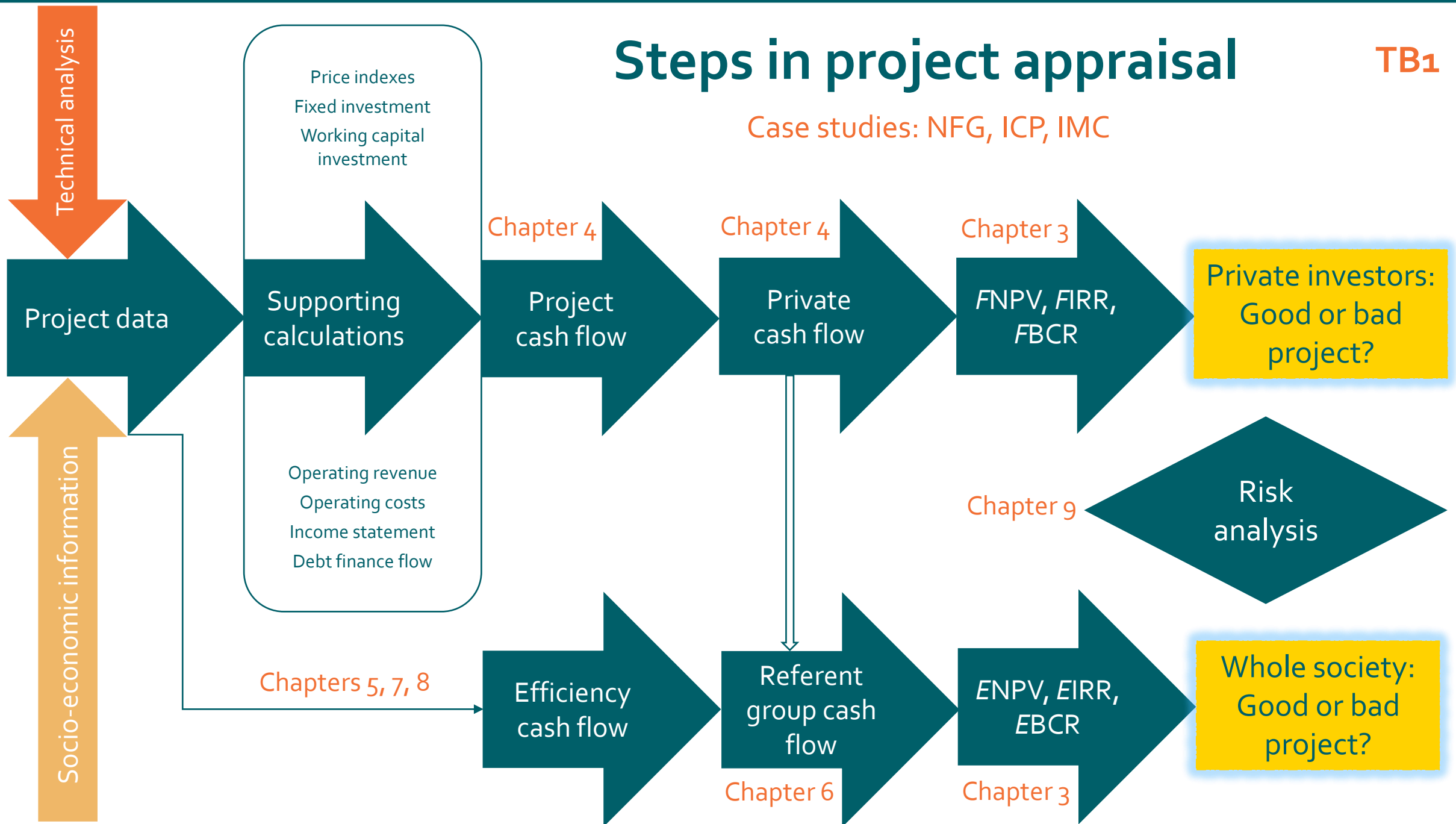
Outline – Module 8

- Introduction
- How to identify referent group net benefits
- Distribution of net benefits: Hypothetical examples
- An illustrative example: NFG project
- An illustrative example: ICP project

Steps in project appraisal

TB1

Case studies: NFG, ICP, IMC



The referent group analysis

An introduction



- **Financial analysis:** Calculate project net benefits at market prices (the Market Analysis) and then disaggregate these values into selected groups – the owners, the banks and the government.
- **Economic analysis:** Calculate project net benefits at efficiency prices, including values of non-marketed goods and services associated with the project.

- **Referent group:** The group of individuals, firms and institutions who have standing in the analysis – i.e. those whose costs and benefits the decision-maker considers relevant to the decision about the project. They are often determined by the discussion with the client who commissioned the project.

- **Referent group analysis:** Calculate the aggregate net benefits to the RG, and then disaggregate them to relevant sub-groups to judge the beneficiaries and losers from a project.
- **Two main reasons of referent group analysis:** (i) Provide information about the distribution of net benefits among sub-groups to the political decision-maker; (ii) Check whether the analysis is correct, or, at least, is internally consistent.

Why referent group analysis is necessary?

- The efficiency analysis is concerned solely with economic efficiency, and ignores the distributional consequences of a project.
- Even if a project is inefficient, it may still be undertaken because of its positive aggregate net benefits to the referent group.
- Even if the aggregate net benefits to the referent group are negative, the decision-maker may still favour the project because it benefits a particular subset of the referent group.

- Efficiency effects resulting from the reallocation of scarce resources.
- Distributional effects determined by externalities as well as by market imperfections, government interventions and distortions.
- The Efficiency Analysis measures the overall net benefits of a project, whereas the **RG Analysis measures the distribution of net benefits among the RG (and non-RG) members.**
- Members of RG experience benefits and costs mainly through external effects and changes in the levels of tax/transfer payments or private rents.

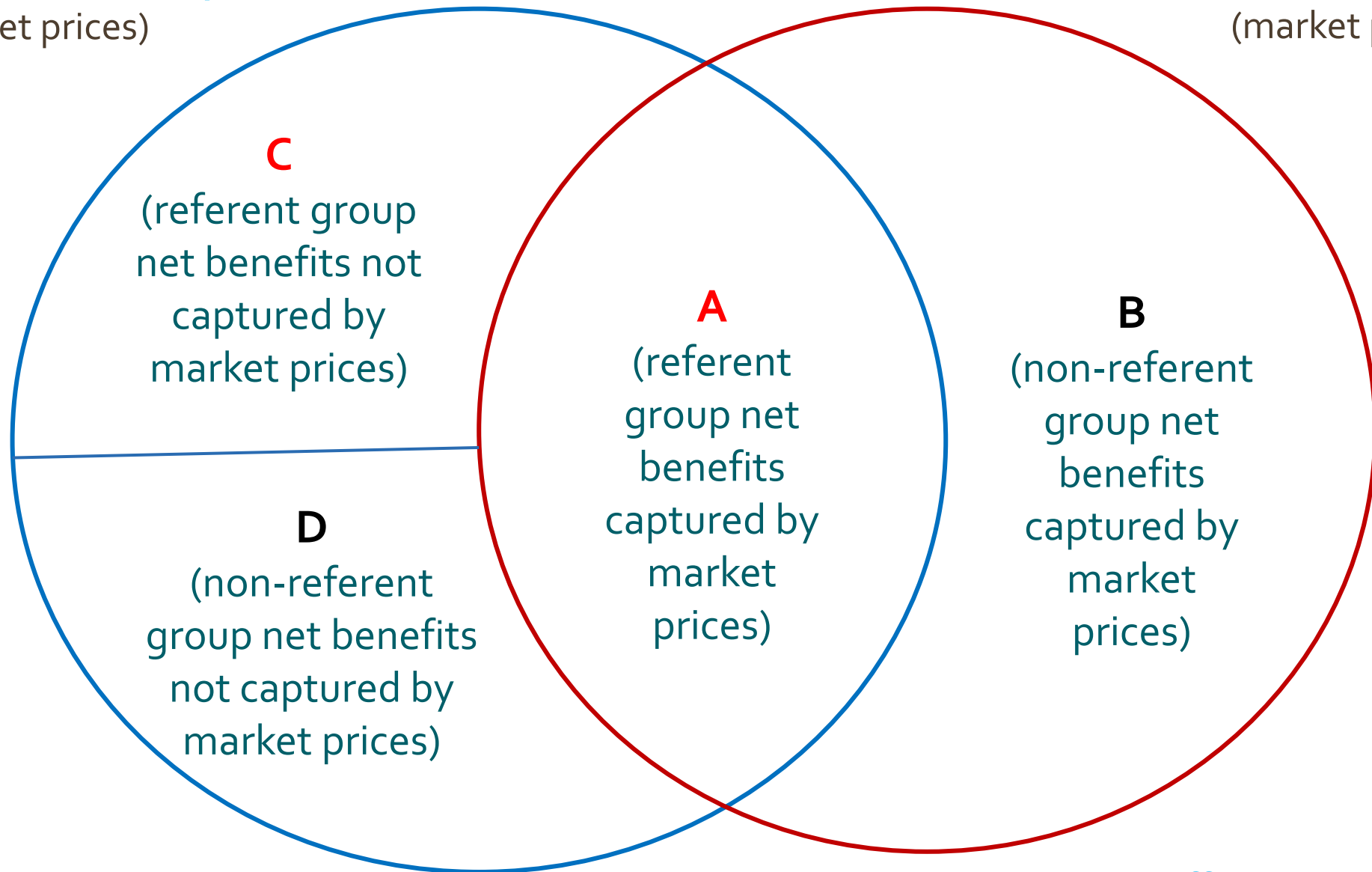
Referent group analysis

How to identify referent group net benefits?



A+B: Market Analysis
(market prices)

B: Private Analysis
(market prices)



A + C: Referent Group Analysis
(efficiency prices)

A + B + C + D: Efficiency Analysis
(efficiency prices)

Referent group net benefit flows

- The total net benefits to RG can be calculated by two methods:
 - **Method 1:** Subtract the net benefits to non-RG members from the efficiency net benefits $[(A + B + C + D - (B + D))]$. This provides us with only the aggregate RG net benefit.
 - **Method 2:** Calculate the benefits and costs to each of the subgroups and add these up. This provides use sufficient information about the distribution of benefits and costs among different subgroups within the RG.
- As a consistency check, the two methods should provide the same aggregate net benefit stream.

Referent group net benefit flows

The project net benefits

Referent group

Non-referent group

Net benefits are measured by market prices

Net benefits are not measured by market prices

Market Analysis

REFERENT GROUP ANALYSIS

Efficiency Analysis

A The net benefits to local government in the form of direct taxes identified by the Market Analysis. This area also includes the net benefits to the domestic banks.

B The net benefits to foreign equity holders and foreign (or interstate) lenders. They are the main non-RG members.

C The net benefits in the form of rents to the landowners in excess of the opportunity cost of land, wages to labour in excess of its opportunity cost, and the costs to downstream users. This area also contains all indirect taxes, subsidies and tariffs paid to or received by government.

D External cost or benefits for residents of other regions or countries not measured by market prices.

Information provided by financial flows

- When tax revenue changes are solely transfers among members of the RG, they net out of the aggregate RG net benefit calculation.
- When changes in tax flows involve transfers between the RG and the rest of the world, the RG experiences a net gain or loss, depending on the direction of the flow.
- The private financing flows are relevant to the construction of the RG cash flow if they transfer net benefits between members and non-members of the RG, or between members of the RG, but are not relevant if they transfer net benefits among non-RG members.

Information provided by shadow prices

	Input	Output
Market price is greater than shadow price	Benefit to owner of the input, e.g. otherwise unemployed labour	Cost to government or public, e.g. loss of tariff revenue, cost of pollution generated by product use
Market price is less than shadow price	Cost to previous user of the input, e.g. monopoly or monopsony firm	Benefit to public, e.g. value of vaccination to those other than the vaccinated

Hypothetical examples

Shadow prices on project inputs



Shadow prices on project inputs

(Example 1: A minimum wage)

e.g.

A domestic firm hires \$100 dollars' worth of labour at the minimum wage and constructs a walking track for which it is paid \$110 by the government. No fee is charged to access the track, and it has a present value of \$120 to consumers. The labour would otherwise be unemployed and would be engaged in non-market activities valued at \$50.

Shadow prices on project inputs

	Referent group	non-Referent group
Net benefits are measured by market prices	A = -\$100 Profit to firm ($110 - 100 = \10) Cost to government ($-\$110$)	B = \$0 None
Net benefits are <u>not</u> measured by market prices	C = \$170 Rent to labour ($100 - 50 = \50) Benefits to users (\$120)	D = \$0 None
Total net benefits	A + C = \$70	B + D = \$0

Shadow prices on project inputs

(Example 2: A maximum price)

e.g.

The state government hires a computer programmer from another state for a month, at a cost of \$200, which is \$50 more than she would normally expect to earn, plus \$100 for accommodation in a rent-controlled apartment with a market value of \$125. The programmer revises some of the government's programs leading to a cost saving of \$400. The Referent Group is defined as the residents of the state.

Shadow prices on project inputs

	Referent group	non-Referent group
Net benefits are measured by market prices	A = \$100 Net benefit to government ($400 - 200 - 100 = \$100$)	B = \$0 None
Net benefits are <u>not</u> measured by market prices	C = -\$25 Loss to potential renter ($100 - 125 = -\$25$)	D = \$50 Rent to labour ($200 - 150 = \50)
Total net benefits	A + C = \$75	B + D = \$50

Shadow prices on project inputs

(Example 3: An indirect tax – a tariff on imports)

e.g.

A domestic firm proposes to import \$100 worth of gadgets for use in a manufacturing project. The \$100 includes the c.i.f. (landed) value of the imported goods of \$75 plus \$25 in tariffs. Other costs are \$200 for skilled labour, and the resulting output will sell on the domestic market for \$400.

Shadow prices on project inputs

	Referent group	non-Referent group
Net benefits are measured by market prices	A = \$100 Profit to firm ($400 - 200 - 100 = \100)	B = \$0 None
Net benefits are <u>not</u> measured by market prices	C = \$25 Tariff revenue to government ($100 - 75 = \$25$)	D = \$0 None
Total net benefits	A + C = \$125	B + D = \$0

Shadow prices on project inputs

(Example 4: A subsidy)

e.g.

The government spends \$100 on improving access to wheat silos, the services of which are supplied free to users. This results in farmers saving \$75 worth of diesel fuel at the subsidized price. The subsidy paid on that quantity of fuel amounts to \$25.

Shadow prices on project inputs

	Referent group	non-Referent group
Net benefits are measured by market prices	A = -\$25 Net benefits to farmers (\$75) Project cost to government (-\$100)	B = \$0 None
Net benefits are <u>not</u> measured by market prices	C = \$25 Government's saving on subsidies (\$25)	D = \$0 None
Total net benefits	A + C = \$0	B + D = \$0

Shadow prices on project inputs

(Example 5: A regulated utility)

e.g.

A foreign company proposes to construct a smelter at a cost of \$100 which will use \$50 worth of power at the current price. The output of the smelter will be worth \$400 on the world market and the company will pay \$100 in income tax. The cost of producing the extra power is \$30.

Shadow prices on project inputs

	Referent group	non-Referent group
Net benefits are measured by market prices	A = \$100 Income tax paid to government (\$100)	B = \$150 Profit to foreign company $(400 - 150 - 100 = \$150)$
Net benefits are <u>not</u> measured by market prices	C = \$20 Extra profit on power supply $(50 - 30 = \$20)$	D = \$0 None
Total net benefits	A + C = \$120	B + D = \$150

Shadow prices on project inputs

(Example 6: Monopoly power)

e.g.

A domestic company is the country's sole producer of cellular phones, which require skilled labour to produce. The government is considering establishing a factory to produce computers, which require the same type of skilled labour. Domestic skilled labour resources are fully employed, and skilled labour cannot be imported. The proposed computer plant would employ \$100 worth of skilled labour and produce \$300 worth of computers.

Shadow prices on project inputs

	Referent group	non-Referent group
Net benefits are measured by market prices	A = \$200 Profit to computer firm (300 – 100 = \$200)	B = \$0 None
Net benefits are <u>not</u> measured by market prices	C = -\$200 Loss to telephone company (-\$200)	D = \$0 None
Total net benefits	A + C = \$0	B + D = \$0

Shadow prices on project inputs

(Example 7: Monopsony power)

e.g.

A domestic company which produces fabricated metal for the export market is the main employer in a remote town. The government proposes establishing a new factory in the town which will use \$100 worth of local labour to produce output of goods valued at \$300.

Shadow prices on project inputs

	Referent group	non-Referent group
Net benefits are measured by market prices	A = \$200 Profit to new company (300 – 100 = \$200)	B = \$0 None
Net benefits are <u>not</u> measured by market prices	C = -\$100 Loss to existing company (-\$100)	D = \$0 None
Total net benefits	A + C = \$100	B + D = \$0

Hypothetical examples

Shadow prices on project outputs



Shadow prices on project outputs

(Example 8: An import-replacing project)

e.g.

A project uses \$200 worth of skilled labour to produce output valued on the domestic market at \$400. Domestic consumers could obtain that same quantity of output from abroad at a cost of \$300 c.i.f. plus \$100 tariff.

Shadow prices on project outputs

	Referent group	non-Referent group
Net benefits are measured by market prices	A = \$200 Net profit to firm (400 – 200 = \$200)	B = \$0 None
Net benefits are <u>not</u> measured by market prices	C = -\$100 Tariff loss to government (-\$100)	D = \$0 None
Total net benefits	A + C = \$100	B + D = \$0

Shadow prices on project outputs

(Example 9: Output of a product subject to subsidy)

e.g.

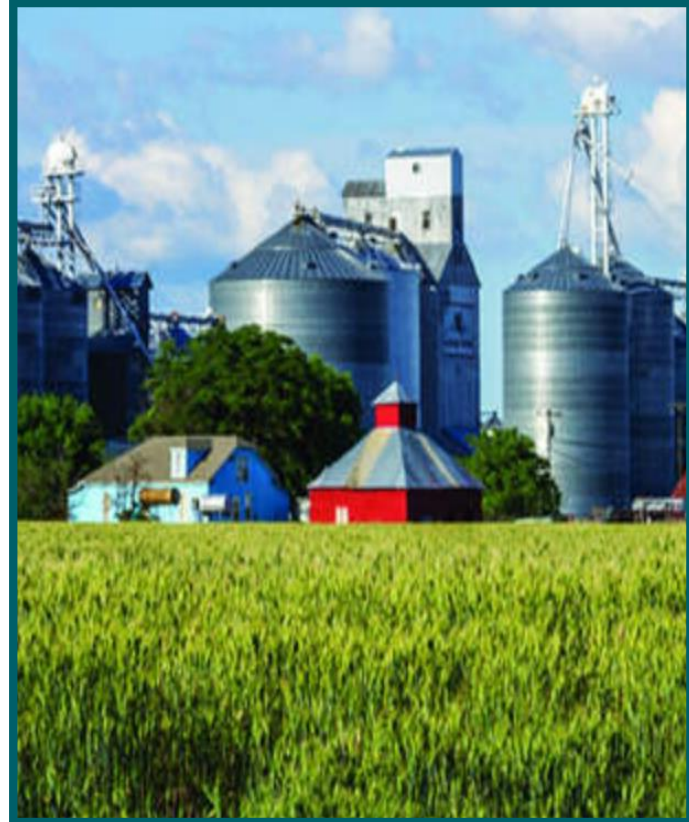
A government project, costing \$100, will increase the supply of irrigation water to sugar producers. While there is no charge for the extra water, farmers will increase the quantity of other inputs used in the production process by \$50 worth at market prices. As a consequence of the increase in water and other inputs the value of the output of sugar, at market prices, will rise by \$125, and farmers will receive an additional \$25 in subsidies.

Shadow prices on project outputs

	Referent group	non-Referent group
Net benefits are measured by market prices	A = \$0 Net profit to farmers ($125 + 25 - 50 = \$100$) Cost of extra water (-\$100)	B = \$0 None
Net benefits are <u>not</u> measured by market prices	C = -\$25 Extra subsidy payment (-\$25)	D = \$0 None
Total net benefits	A + C = -\$25	B + D = \$0

Hypothetical examples

Corrective taxation



Corrective taxation

(Example 10: Cost of fuel subject to a corrective tax)

e.g.

The government spends \$100 on improving port facilities to provide sawmills with access to a new market. This results in sawmills receiving an extra \$200 for their timber but spending an extra \$125 on diesel fuel to transport their product to the port. The market value of the extra fuel includes \$25 tax which is imposed to discourage consumption of fuel because of its related health effects.

Corrective taxation

	Referent group	non-Referent group
Net benefits are measured by market prices	A = -\$25 Net benefits to sawmills ($200 - 125 = \$75$) Project cost to government (-\$100)	B = \$0 None
Net benefits are <u>not</u> measured by market prices	C = \$0 Government gain in tax revenue (\$25) Rise in community health cost (-\$25)	D = \$0 None
Total net benefits	A + C = -\$25	B + D = \$0

Corrective taxation

(Example 11: Cost of fuel subject to a corrective subsidy)

e.g.

Now suppose that the extra fuel used in Example 6.10 was biofuel – such as a blend of diesel and ethanol – which results in lower emissions of particulates and lower associated health costs as compared with regular fuel. We know from the above example that the production cost of the diesel saved is \$100 – the market price less the \$25 tax. Suppose that the production cost of the biofuel is \$110 and that a corrective subsidy of \$10 is paid to producers to render the biofuel competitive with diesel at their tax-inclusive prices. If both the tax and the subsidy are corrective, we can infer that, while burning the diesel fuel imposes a health cost of \$25, burning the biofuel has a \$15 external cost. We can redo the net benefit calculations on the assumption that the fuel saved was biofuel.

Corrective taxation

	Referent group	non-Referent group
Net benefits are measured by market prices	A = -\$25 Net benefit to farmers (200 - 125 = \$75) Project cost to government (-\$100)	B = \$0 None
Net benefits are <u>not</u> measured by market prices	C = \$0 Extra fuel tax revenue (\$25) Extra biofuel subsidy (-\$10) Extra health costs (-\$15)	D = \$0 None
Total net benefits	A + C = -\$25	B + D = \$0

Corrective taxation

(Example 12: Production of cigarettes subject to a corrective tax)

e.g.

A private firm plans to produce cigarettes to satisfy additional demand. The market value of its output will be \$250, which includes indirect tax of \$50. The cost of the extra production is \$150 and the firm will pay income tax of \$25.

Corrective taxation

	Referent group	non-Referent group
Net benefits are measured by market prices	A = \$50 Net profit to firm ($200 - 150 - 25 = \25) Income tax revenue to government (\$25)	B = \$0 None
Net benefits are <u>not</u> measured by market prices	C = -\$0 Extra health costs (-\$50) Corrective tax (\$50)	D = \$0 None
Total net benefits	A + C = \$50	B + D = \$0

Corrective taxation

(Example 13: A subsidy on production of vaccine)

e.g.

A pharmaceutical company plans to supply additional influenza vaccinations worth \$150 at the market price. In addition, it will receive a \$50 subsidy from government. The cost of the extra quantity produced is \$175.

Corrective taxation

	Referent group	non-Referent group
Net benefits are measured by market prices	A = -\$25 Net profit to firm ($150 + 50 - 175 = \25) Subsidy from government (-\$50)	B = \$0 None
Net benefits are <u>not</u> measured by market prices	C = \$50 Reduced health costs (\$50)	D = \$0 None
Total net benefits	A + C = \$25	B + D = \$0

Hypothetical examples

Further examples



Further examples

(Example 14: Where the market and efficiency net benefits are identical)

e.g.

A local firm engages a foreign contractor to assist with the upgrade of an existing plant. The foreign contractor is paid a management fee at the going international rate of \$300. The firm spends a further \$500 on new technology, purchased abroad at the current world price. The result is an increase in the firm's output of exports, sold at the going world price, equal in value to \$1200. The firm pays \$100 additional taxes to government and shares the remaining profit 50:50 with the foreign contractor, who is not considered as part of the Referent Group.

Further examples

	Referent group	non-Referent group
Net benefits are measured by market prices	A = \$250 Profit to local firm ($1200 - 800 - 100 - 150 = \150) Direct taxes to government (\$100)	B = \$150 Profit to foreign contractor ($50\% \times (1200 - 800 - 100 = \$150)$)
Net benefits are <u>not</u> measured by market prices	C = \$0 None	D = \$0 None
Total net benefits	A + C = \$250	B + D = \$150

Further examples

(Example 15: Where there are market and non-market non-RG net benefits)

e.g.

A foreign-owned agricultural enterprise invests \$420 in an irrigation project, including \$220 for a new pump from an inter-state supplier which cost \$200 to supply. It employs some local casual labour costing \$100, and an inter-state irrigation consultant costing \$150 and pays \$30 in income tax to the state government. The value of its output of fruit at domestic prices increases by \$800, but the project causes \$10 in lost output in a neighbouring state due to reduced water flow to downstream users. The opportunity cost of the local labour is \$50 and the opportunity cost of the inter-state consultant is \$100. The value of the enterprise's increased output at world prices is \$750, which is the value at domestic prices less the tariff on imports of fruit. Assume that all foreign and inter-state parties are non-Referent Group members, and that the additional output of fruit replaces imports.

Further examples

Referent group

non-Referent group

Net benefits are measured by market prices	A = \$30 Taxes to government (\$30)	B = \$100 Profit to agricultural enterprise ($800 - 420 - 100 - 150 - 30 = \100)
Net benefits are <u>not</u> measured by market prices	C = \$0 Rent to local labour ($100 - 50 = \50) Loss of tariff to government ($750 - 800 = -\$50$)	D = \$60 Profit to interstate pump supplier ($220 - 200 = \$20$) Rent to interstate consultant ($150 - 100 = \$50$) Loss to downstream water users ($-\$10$)
Total net benefits	A + C = \$30	B + D = \$160

An illustrative examples

NFG Project, Campbell & Brown (2015, 169-170)



REFERENT GROUP CASH FLOW

	2015	2016	2017	2018	2019	2020	2029	2030	2031	2032	2033	2034	2035
Year	0	1	2	3	4	5	14	15	16	17	18	19	20
RG NET BENEFITS CALCULATION													
Method 1: Efficiency minus non-RG cash flow (NFG + Banks)	51.4	-40.7	-31.9	-30.4	-29.0	-28.4	-9.5	-9.5	-9.5	-9.5	-9.5	-9.5	3.2
Method 2: Distribution by sub-groups													
Government	51.4	-42.6	-25.8	-26.3	-26.8	-26.2	-7.3	-7.3	-7.3	-7.3	-7.3	-7.3	5.4
- Import duties	51.4	1.7	0.8	1.2	1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.6	-4.9
- Indirect taxes		0.1	0.1	0.1	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.1
- Company taxes		-16.0	-8.3	0.0	8.3	8.9	27.8	27.8	27.8	27.8	27.8	27.8	27.8
- Subsidies		-28.4	-15.4	-27.6	-36.8	-36.8	-36.8	-36.8	-36.8	-36.8	-36.8	-36.8	-17.6
Landowners' rent		0.8	1.5	2.3	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
Labour rent		1.2	2.4	3.6	4.8	4.8	4.8	4.8	4.8	4.8	4.8	4.8	4.8
Downstream users		0.0	-10.0	-10.0	-10.0	-10.0	-10.0	-10.0	-10.0	-10.0	-10.0	-10.0	-10.0
Referent group cash flow	51.4	-40.7	-31.9	-30.4	-29.0	-28.4	-9.5	-9.5	-9.5	-9.5	-9.5	-9.5	3.2
Check for consistency	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Social discount rates		5%	10%	15%
Government		-173.2	-124.1	-91.5
Direct taxes		176.3	92.6	49.5
Indirect taxes		1.8	1.2	0.8
Import duties		67.5	62.9	60.0
Subsidies		-418.8	-280.7	-201.9
Landowners' rent		33.2	21.7	15.2
Labour rent		53.2	34.7	24.3
Downstream users		-115.1	-76.0	-53.9
Referent Group		-201.9	-143.7	-105.9
Non-Referent Group		433.3	7.8	-227.9

NPV of each
RG stakeholder

Method 1: $(A + B + C + D) - (B + D)$

Phương pháp 2: Sum of
all sub-groups' net
benefits

Net benefit of NFG (Private Analysis) = B₁

Net benefit of project (Market Analysis) = A + B = A + B₁ + B₂

Suất chiết khấu	5%	10%	15%
NPV(Dự án)	609.56	100.40	-178.35
NPV(TIPV)	433.26	7.79	-227.89
NPV(EPV)	483.33	196.37	59.62
NPV(Ngân hàng)	-50.07	-188.58	-287.51
NPV(Thuế trực thu)	176.30	92.61	49.54
Kiểm tra	609.56	100.40	-178.35

Net benefit of government (Income taxes) = A

Net benefit of lenders (Private Analysis) = B₂

NFG project

	Referent Group	Non-Referent Group	NPV
Net benefits are measured by market prices	A = \$92.6 Direct taxes (+\$92.6)	B = +\$7.8 NFG firm (+\$196.4) Banks (-\$188.6)	A + B = +\$100.4
Net benefits are <u>not</u> measured by market prices	C = -\$236.3 Indirect taxes (-\$216.7) Landowners' rent (+\$21.7) Labour rent (+\$34.7) Downstream users (-\$76)	D = \$0 None	C + D = -\$236.3
Total net benefits	A + C = -\$143.7	B + D = +\$7.8	A + B + C + D = -\$135.9

An illustrative examples

ICP Project, Campbell & Brown (2016, chapters 6, 13)



	A	B	C	D	E	F	G	H	I	J	K	L
1	OPTION 1: BANGKOK - NO CONCESSIONS											
2												
70	Efficiency Analysis											
71												
72		1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009
73	INPUTS											
74	Investment Costs											
75	Fixed Investment	-253.8										
76	Working Capital		-63.5									63.5
77	TOTAL	-253.8	-63.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	63.5
78												
79	Operating Costs											
80	Raw Materials		-106.9	-213.8	-213.8	-213.8	-213.8	-213.8	-213.8	-213.8	-213.8	-213.8
81	Labour		-40.5	-40.5	-40.5	-40.5	-40.5	-40.5	-40.5	-40.5	-40.5	-40.5
82	Fuel, Water		-5.4	-10.8	-10.8	-10.8	-10.8	-10.8	-10.8	-10.8	-10.8	-10.8
83	Spare Parts		-21.3	-21.3	-21.3	-21.3	-21.3	-21.3	-21.3	-21.3	-21.3	-21.3
84	Insurance and Rent		-3.0	-3.0	-3.0	-3.0	-3.0	-3.0	-3.0	-3.0	-3.0	-3.0
85	TOTAL		-177.2	-289.5	-289.5	-289.5	-289.5	-289.5	-289.5	-289.5	-289.5	-289.5
86												
87	OUTPUTS											
88	Revenues											
89	Cost Savings		127.6	255.2	255.2	255.2	255.2	255.2	255.2	255.2	255.2	255.2
90	Yarn Sales		63.8	127.6	127.6	127.6	127.6	127.6	127.6	127.6	127.6	127.6
91	TOTAL		191.4	382.8	382.8	382.8	382.8	382.8	382.8	382.8	382.8	382.8
92												
93	EFFICIENCY ANALYSIS	-253.8	-49.3	93.3	93.3	93.3	93.3	93.3	93.3	93.3	93.3	156.8
94												
95	Discount Rates	4%	8%	12%	16%	20%						
96	NPV	408.9	269.8	166.6	88.7	28.9						
97												
98	IRR	22%										

	A	B	C	D	E	F	G	H	I	J	K	L
106	Referent Group Analysis											
107												
108	METHOD A:											
109		1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009
110	Domestic Bank											
111	Overdraft		-60.0									60.0
112	Interest from Overdraft			6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0
113	Thai Government											
114	Lost Import Duties		-18.0	-36.0	-36.0	-36.0	-36.0	-36.0	-36.0	-36.0	-36.0	-36.0
115	Spindles I.Duty	8.8										
116	Equipment I.Duty	1.9										
117	WCap Raw Materials I.Duty		5.3									-5.3
118	WCap Spare Parts I.Duty		0.5									-0.5
119	Imported Cotton I.Duty		10.7	21.4	21.4	21.4	21.4	21.4	21.4	21.4	21.4	21.4
120	Spare Parts Duty		1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
121	Profits Tax		-13.3	27.1	28.3	29.6	30.9	32.4	32.4	32.4	32.4	32.4
122	Domestic Labour											
123	Initial Labour	18.0										
124	Ongoing Labour		3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
125	Domestic Electricity Corp											
126	Electricity		1.8	3.6	3.6	3.6	3.6	3.6	3.6	3.6	3.6	3.6
127	Domestic Insurance Co											
128	Insurance		3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
129	TOTAL	28.7	-65.9	29.2	30.3	31.6	33.0	34.5	34.5	34.5	34.5	88.6
130												
131	METHOD B:	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009
132												
133	Efficiency Analysis	-253.8	-49.3	93.3	93.3	93.3	93.3	93.3	93.3	93.3	93.3	156.8
134	Less Non-Referent Grp											
135	ICP	-122.4	-23.4	24.1	22.9	21.6	20.3	58.9	58.9	58.9	58.9	68.2
136	Foreign Credit - Principal	-160.1	27.3	29.5	31.8	34.4	37.1					
137	Foreign Credit - Interest		12.8	10.6	8.3	5.7	3.0					
138	TOTAL	28.7	-65.9	29.2	30.3	31.6	33.0	34.5	34.5	34.5	34.5	88.6
139												
140	Discount Rates	4%	8%	12%	16%	20%						
141	NPV	236.0	181.1	141.4	112.1	90.2						
142												
143	IRR	N/A										

ICP Project (r = 10%/year)

	Referent Group	Non-Referent Group	NPV
Net benefits are measured by market prices	A = \$147.2 Direct taxes (+\$147.2) Domestic bank (\$0)	B = +\$54.8 ICP company (+\$62.9) Foreign bank (-\$8.1)	A + B = +\$202.0
Net benefits are <u>not</u> measured by market prices	C = +\$12.5 Indirect taxes (-\$62.9) Labour rent (+\$36.4) Power supplier (+\$20.5) Insurance company (+\$18.4)	D = \$0 None	C + D = \$12.5
Total net benefits	A + C = \$159.7	B + D = +\$54.8	A + B + C + D = \$214.5

Discount Rates (for NPV): Efficiency analysis		4%	8%	12%	16%	20%	RANK	IRR	RANK
BANGKOK	Option 1	408.9	269.8	166.6	88.7	28.9	2	22%	2
	Option 2	408.9	269.8	166.6	88.7	28.9		22%	
	Option 3	408.9	269.8	166.6	88.7	28.9		22%	
SOUTHERN THAILAND	Option 4	408.1	271.4	170.1	93.6	39.4	1	23%	1
	Option 5	408.1	271.4	170.1	93.6	39.4		23%	
	Option 6	408.1	271.4	170.1	93.6	39.4		23%	
	Option 7	408.1	271.4	170.1	93.6	39.4		23%	
Discount Rates (for NPV): Referent group analysis		4%	8%	12%	16%	20%	RANK (not 20%)	IRR	
BANGKOK	Option 1	236.0	181.1	141.4	112.1	90.2	2	N/A	
	Option 2	154.5	114.3	85.7	65.0	49.8	4	N/A	
	Option 3	29.3	16.9	8.9	3.7	0.5	6	N/A	
SOUTHERN THAILAND	Option 4	270.1	217.7	179.9	152.1	131.4	1	N/A	
	Option 5	188.5	150.9	124.3	105.0	91.0	3	N/A	
	Option 6	92.5	77.5	67.5	60.9	56.4	5	N/A	
	Option 7	-70.7	-56.1	-43.8	-33.3	-24.3	7	N/A	

Referent group analysis

Q&A section

